Kalos Developer Kit Carrier Board Specification

DX-3000-D

Specification

Document History

Author	Version	Date	Description of Change
HaoNing	1.0	2026.1.2	Document initial
HaoNing	1.1	2026.2.6	Added : Device ordering information
HaoNing	2.0	2026.4.29	Updated: 1. Carrier Board Mechanical Dimensions 2. J12/J19/J25 headers pin definition and change to 2.0mm pitch 3. Changed J3 LCD MIPI Connector Pin definition and Part NO. 4. Updated and corrected technical parameters. 5. Changed CAN1/2 ports from logic to HL bus ports.
HaoNing	2.0.1	2026.6.3	Updated: 1. Power diagram 2. Supply Current Capabilities table
HaoNing	2.0.2	2026.6.29	Updated and corrected technical parameters.
Shichenkang	2.0.3	2026.6.29	1. Removed duplicated 'Carrier Board' in table captions; 2. converted full-width punctuation to half-width in English text; 3. removed zero-width characters.
Shichenkang	2.0.4	2026.6.29	1. Reworded USB feature list: moved 2x USB Type-C to a top-level item so the USB 3.0 hub's 4x count applies only to the 4x USB Type-A ports.
HaoNing	2.0.5	2026.9.9	Updated devkit operating temperature range

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1 Introduction

This specification contains recommendations and guidelines for engineers to follow to create modules for the expansion connectors on the Kalos Module carrier board. As well as understand the Capabilities of the other dedicated interface connectors and associated power solutions on the platform.

1.1 Kalos Carrier Board Feature List

The following is a list of features for the carrier board.

- Connection to Kalos Module
  - 260 pin SO-DIMM Connector
- USB Ports
  - USB 3.0 Hub to 4x
    - 4x USB Type-A USB3.0 (host only)
  - 2x USB Type-C : 1x USB2.0(Device) ,1x USB-C (UART debug only)
- Wired network
  - Gigabit Ethernet 2x RJ45 connectors with LEDs
- Display video Ports
  - 1x HDMI 1.4
  - 1x 22-position flex connectors
    - DSI , D-PHY(V1.2) Support 4 data-lanes per port,2.5Gbps per lane
- Camera connectors
  - 2x 22-position flex connectors
    - CSI , D-PHY(V2.1) 2 Lanes or 4 Lanes,2.5Gbps per lane
    - Camera CLK, I2C, and control
- M.2 Key M connector
  - PCIe (Gen3) x4 Lane, control RC or EP Mode
- Expansion header
  - 40 + 20 expansion header
    - 3xCAN FD/2xSPI/2xUART/4xPWM/2xI2S/8xI2C/5xGPIO/3V3/5V/1V8
- UI and indicators
  - Button header: Power, reset, force recovery,
  - LEDs: Power
- Power and Miscellaneous
  - Fan connector: 5V/12V, PWM
  - Power DC Jack: 9-24 V (12V supply provided)
  - Main 5.0 V
  - Main 3.3V
- Developer kit operating temperature range
  - 0~45℃

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1.2 Kalos Carrier Board Block Diagram

Figure 1 to Figure 5 show the block diagram and various placement views for Kalos core board and the carrier board.

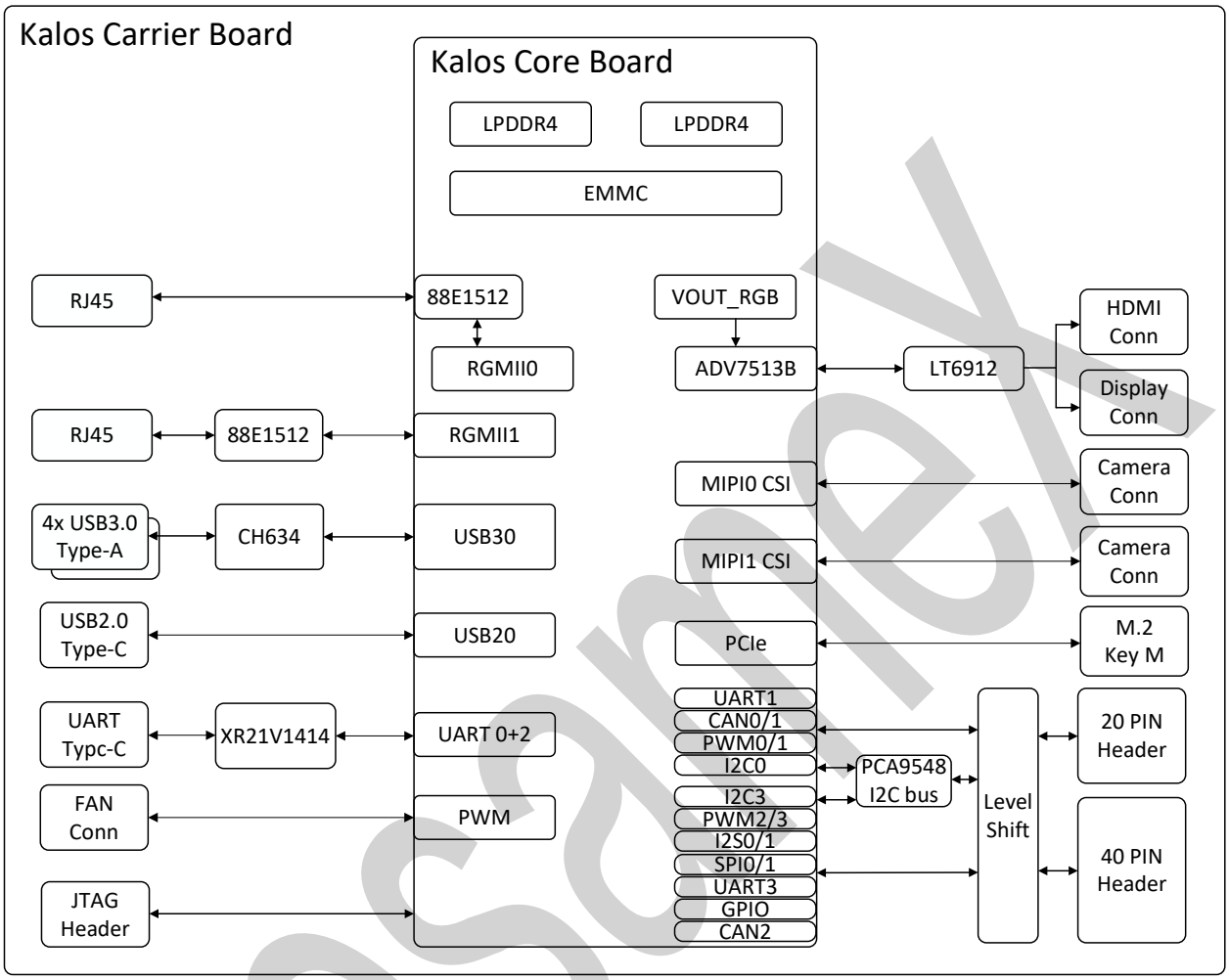


Figure 1. Kalos Module Core&Carrier Board Block diagram

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2 Mechanical Specification

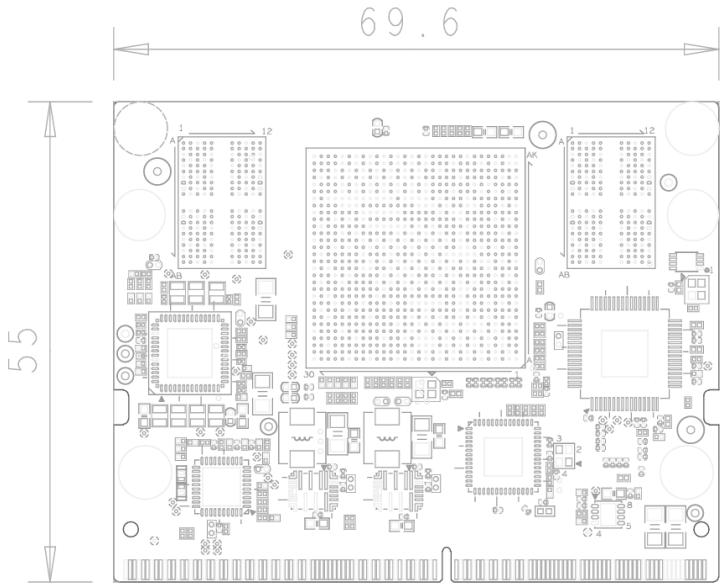


Figure 2. Kalos Module Core Board Mechanical Dimensions

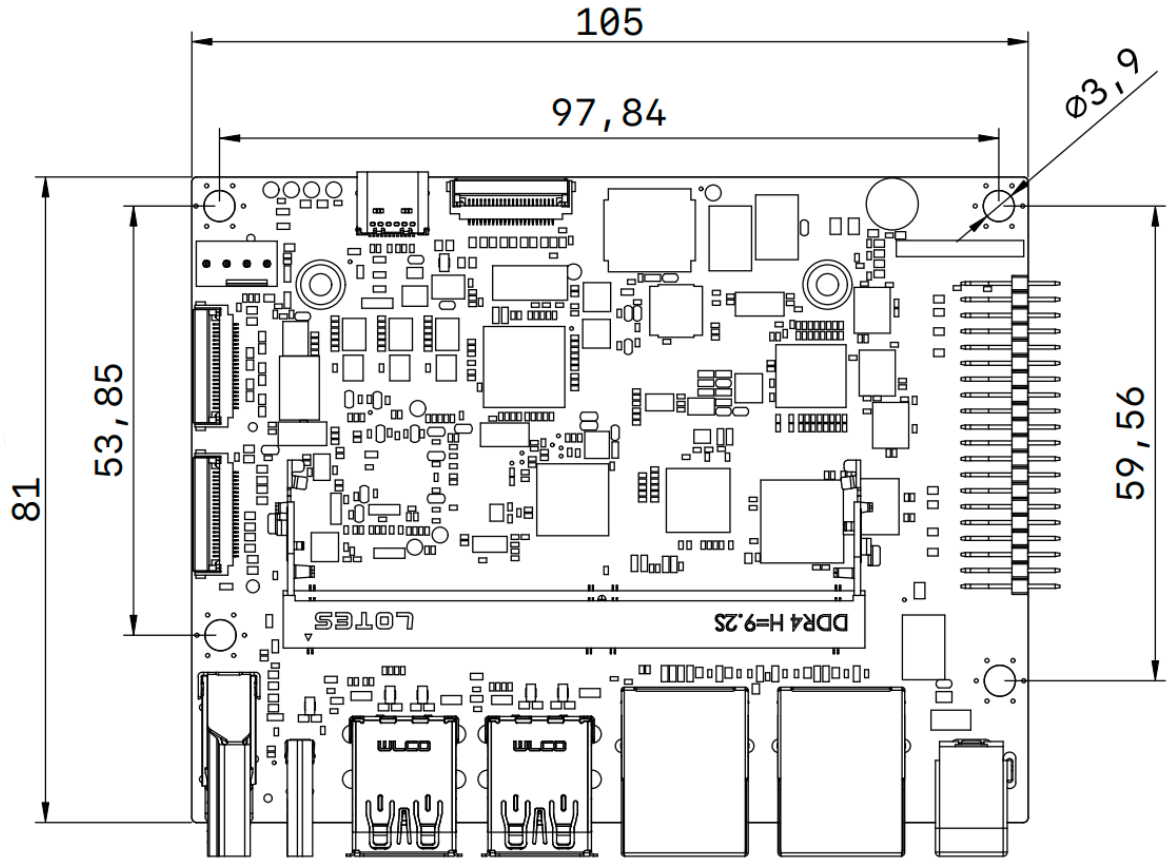


Figure 3. Kalos Carrier Board Mechanical Dimensions

3 Carrier Board Connectors Placement

J2	Kalos Core Board Conn.(SODIMM,260-pin)	**J5**	USB Type-C
J6	USB Type A Dual Stacked Connector	**J9**	USB Type-C
J16	Power Jack	**J18**	USB Type A Dual Stacked Connector
J23	HDMI Connector	**J25**	8-pin JTAG Header (2 mm pitch)
J4	RJ45 Ethernet Socket	**J3**	Display Connector(22 pin, 0.5 mm pitch)
J12	40-pin Expansion Header (2x20, 2 mm pitch)	**J15**	RJ45 Ethernet Socket
J19	20-pin Expansion Header (1x20, 2 mm pitch)	**J21**	Camera1 Connector(22 pin, 0.5 mm pitch)
J20	Camera0 Connector(22 pin, 0.5 mm pitch)	**J24**	M.2 Key M slot
J32	Fan Header		

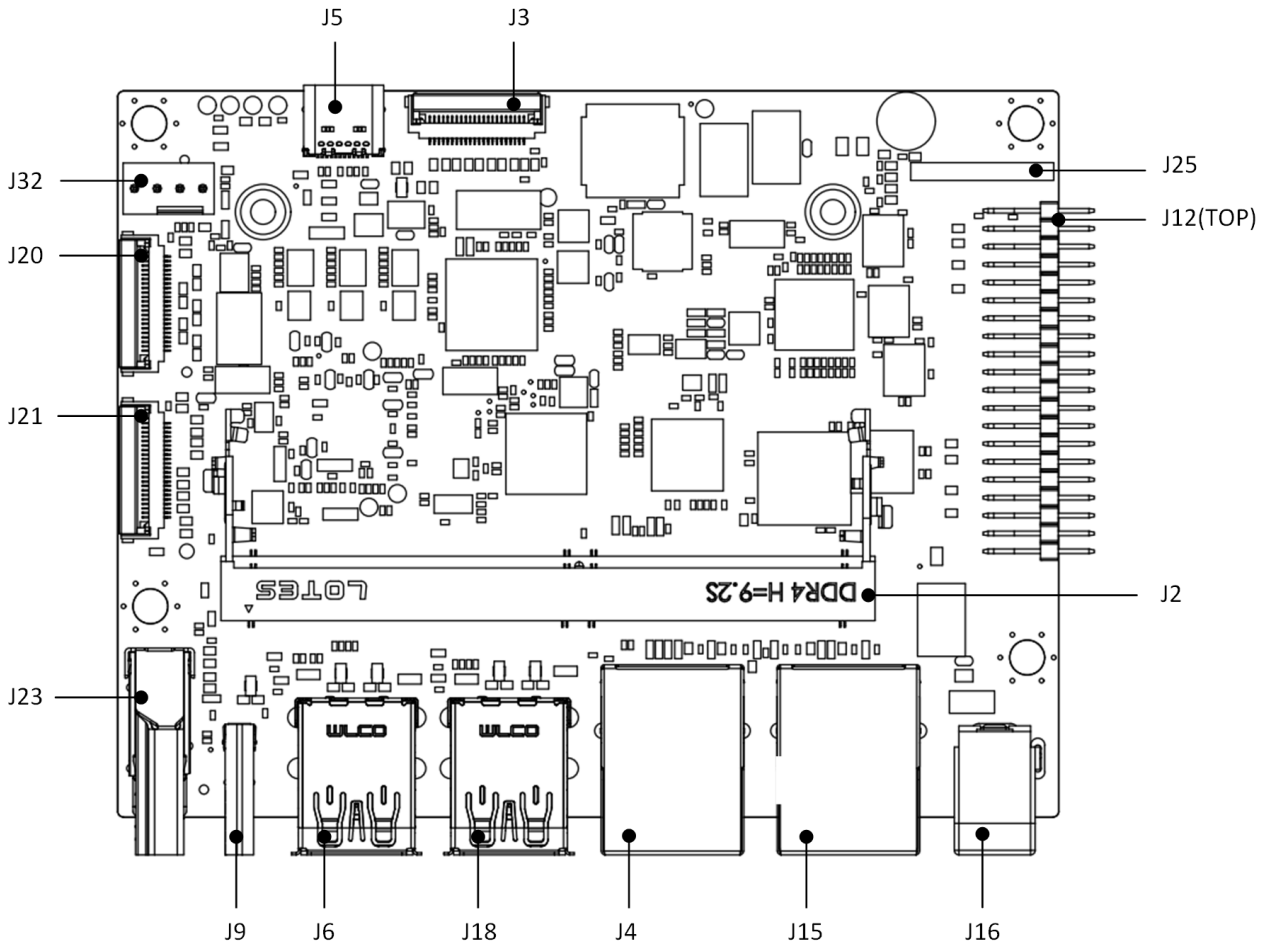


Figure 4. Kalos Carrier Board TOP side

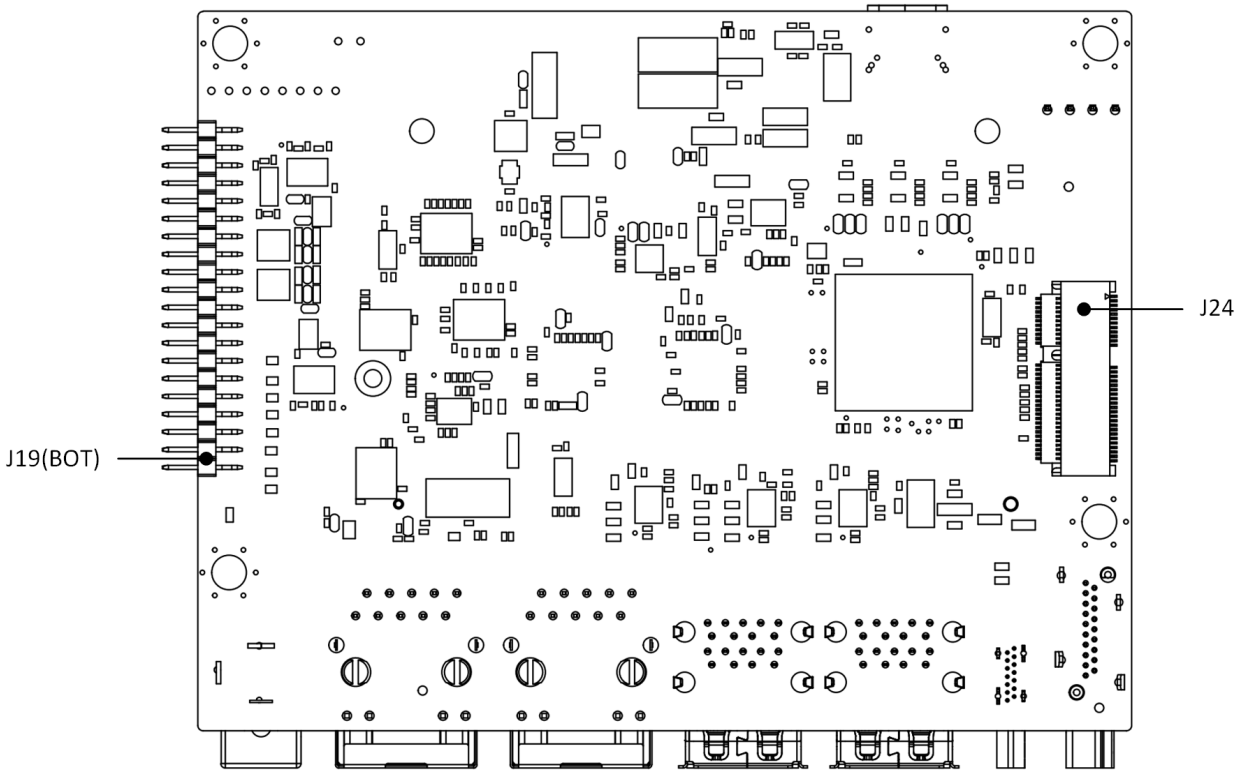


Figure 5. Kalos Carrier Board BOT side

4 Board Standard Connectors

Kalos carrier board provides several connectors with industry standard pinouts to support additional functions. This includes:

- *USB 2.0: 1x Type C connector (Device Only)*
- *UART Debug: 1x Type C connector*
- *USB 3.0: 4 x Type A stacked connectors*
- *Gigabit Ethernet: 2 x RJ45 connector*
- *HDMI*
- *M.2 Key M socket (4-lane PCIe)*

4.1 USB Port

The carrier board supports six USB connectors. J5 USB Type-C connector supports device-only mode and is dedicated solely for downloads. In addition, there are two, dual stacked USB 3.0 Type A connectors (J6 and J18). Each Type A connector supports host mode only.

- *4x USB Type-A USB3.0*
- *2x USB Type-C:*
  - *1x USB2.0 for device only (J5)*
  - *1x USB-C for UART debug only (J9)*

Table1-1 USB Type-A Connector Pin Descriptions – J6

Pin NO.	Pin Name	——	Usage/Description	Type/Dir

Type A 1#				
L1	P5V_USB_A1		VBUS Supply	Power
L2	USB1_DM		USB 2.0 Data from hub	Bidir
L3	USB1_DP			
L4	GND		Ground	Ground
L5	USB_SSRN1		USB3.0 Receiver Data from Hub	Input
L6	USB_SSRP1			
L7	GND		Ground	Ground
L8	USB_SSTN1		USB3.0 Transmit Data from Hub	Output
L9	USB_SSTP1			
Type A 2#				
T1	P5V_USB_A2		VBUS Supply	Power
T2	USB2_DM		USB 2.0 Data from hub	Bidir
T3	USB2_DP			
T4	GND		Ground	Ground
T5	USB_SSRN2		USB3.0 Receiver Data from Hub	Input
T6	USB_SSRP2			
T7	GND		Ground	Ground
T8	USB_SSTN2		USB3.0 Transmit Data from Hub	Output
T9	USB_SSTP2			

Table 1-2 USB Type-A Connector Pin Descriptions – J18

Pin NO.	**Pin Name**	——	**Usage/Description**	**Type/Dir**
Type A 1#				
1	P5V_USB_A3		VBUS Supply	Power
2	USB3_DM		USB 2.0 Data from hub	Bidir
3	USB3_DP			
4	GND		Ground	Ground
5	USB_SSRN3		USB3.0 Receiver Data from Hub	Input
6	USB_SSRP3			
7	GND		Ground	Ground
8	USB_SSTN3		USB3.0 Transmit Data from Hub	Output
9	USB_SSTP3			
Type A 2#				
10	P5V_USB_A4		VBUS Supply	Power
11	USB4_DM		USB 2.0 Data from hub	Bidir
12	USB4_DP			
13	GND		Ground	Ground
14	USB_SSRN4		USB3.0 Receiver Data from Hub	Input
15	USB_SSRP4			
16	GND		Ground	Ground
17	USB_SSTN4		USB3.0 Transmit Data from Hub	Output
18	USB_SSTP4			

Table 2 USB Type-C Connector Pin Descriptions – J5

Pin NO.	**Pin Name**	——	**Usage/Description**	**Type/Dir**
A1	GND			

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/				
/				
A4	P5V_USB_C1		VBUS Supply	Power
A5	100K TO GND		CC pin pull down	Output
A6	USB_C1_DP		USB 2.0 Data from SOC	Bidir
A7	USB_C1_DM			
A8	GND			
A9	P5V_USB_C1		VBUS Supply	Power
/				
/				
A12	GND			
B12	GND			
/				
/				
B9	P5V_USB_C1		VBUS Supply	Power
B8	GND			
B7	TYPEC1_USBM		USB_C1_DM	Bidir
B6	TYPEC1_USBP		USB_C1_DP	
B5	100K TO GND			Output
B4	P5V_USB_C1		BUS Supply	Power
/				
/				
B1	GND			

Table 3 USB Type-C Connector Pin Descriptions – J9

Pin NO.	**Pin Name**	——	**Usage/Description**	**Type/Dir**
A1	GND			
/				
/				
A4	P5V_USB_C2	USB0_VBUS_CON	VBUS Supply	Power
A5	100K TO GND		CC pin pull down	Output
A6	USB_C2_DP		From UART to USB	Bidir
A7	USB_C2_DM			
A8	GND			
A9	P5V_USB_C2		VBUS Supply	Power
/				
/				
A12	GND			
B12	GND			
/				
/				
B9	P5V_USB_C2		VBUS Supply	Power
B8	GND			
B7	TYPEC2_USBM		USB_C2_DM	Bidir
B6	TYPEC2_USBP		USB_C2_DP	
B5	100K TO GND			Output
B4	P5V_USB_C2		BUS Supply	Power

Sesame

/				
/				
B1	GND			

4.2 GbE Ethernet

The carrier board supports two RJ45 connectors for gigabit ethernet (J4/J15).

Table 4 Ethernet RJ45 Pin Description – J4

Pin NO.	Pin Name	——	Usage/Description	Type/Dir
1	CT1			
2	GBE_MDI0_P			Bidir
3	GBE_MDI0_N			
4	GBE_MDI1_P			
5	GBE_MDI2_P			
6	GBE_MDI2_N			
7	GBE_MDI1_N			
8	GBE_MDI3_P			
9	GBE_MDI3_N			
10	GND			
11	VCC3_MGT		L0_G_LED+	
12	GBE_LED_ACT		L0_G_LED-	
13	GBE_LED_LINK		R_Y_LED-	
14	VCC3_MGT		R_Y_LED+	

Table 5 Ethernet RJ45 Pin Description – J15

Pin NO.	Pin Name	——	Usage/Description	Type/Dir
1	CT2			
2	GBE1_MDI0_P			Bidir
3	GBE1_MDI0_N			
4	GBE1_MDI1_P			
5	GBE1_MDI2_P			
6	GBE1_MDI2_N			
7	GBE1_MDI1_N			
8	GBE1_MDI3_P			
9	GBE1_MDI3_N			
10	GND			
11	VCC3_MGT		L0_G_LED+	
12	GBE_LED_ACT		L0_G_LED-	
13	GBE_LED_LINK		R_Y_LED-	
14	VCC3_MGT		R_Y_LED+	

4.3 HDMI

The carrier board includes one HDMI 1.4 Port(J23). Output which provides sufficient bandwidth for 1080p@60fps video.

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Table 6 HDMI Port Description – J23

Pin NO.	Pin Name	Usage/Description	Pin NO.	Pin Name	Usage/Description
1	HTX_D2+	TMDS Data2	2	GND	
3	HTX_D2-		4	HTX_D1+	TMDS Data1
5	GND		6	HTX_D1-	
7	HTX_D0+	TMDS Data0	8	GND	
9	HTX_D0-		10	HTX_CLK+	TMDS clock
11	GND		12	HTX_CLK-	
13	HTX_CEC	Consumer Electronic control	14	/	
15	TX_SCL		16	TX_SDA	
17	GND		18	TX_5V	Power Supply
19	TX_HPD	HDMI Hotplug			

4.4 M.2 Key M Slot

The carrier board includes one M.2, Key M Slots for NVMe storage (J24). The M.2, Key M slot supports PCIe (x4), Gen3.

Table 7 M.2 Key M Slot Pin Description – J24

Pin NO.	Pin Name	Usage/Description	Type/Dir	Pin NO.	Pin Name	Usage/Description	Type/Dir
1	GND			2	P3V3_M2M		
3	GND			4	P3V3_M2M		
5	A_PCIE1_RX1_N			6	/		
7	A_PCIE1_RX1_P			8	/		
9	GND			10	/		
11	A_PCIE1_TX1_N			12	P3V3_M2M		
13	A_PCIE1_TX1_P			14	P3V3_M2M		
15	GND			16	P3V3_M2M		
17	A_PCIE1_RX0_N			18	P3V3_M2M		
19	A_PCIE1_RX0_P			20	A_PCIE1_CLKREQ		
21	GND			22	A_PCIE1_RST		
23	A_PCIE1_TX0_N			24	W_DISABLE1_CTRL		
25	A_PCIE1_TX0_P			26	BT_M2_EN		
27	GND			28	BT_M2_WAKE_AP		
29	A_PCIE0_RX1_N	PCIE lane1 Recevie	IN	30	/		
31	A_PCIE0_RX1_P			32	/		
33	GND			34	M2_E_USB2_DP		
35	A_PCIE0_TX1_N	PCIE lane1 Transmit	OUT	36	M2_E_USB2_DN		
37	A_PCIE0_TX1_P			38	/		
39	GND			40	/		
41	A_PCIE0_RX0_N	PCIE lane0 Receive	IN	42	/		
43	A_PCIE0_RX0_P			44	/		
45	GND			46	/		
47	A_PCIE0_TX0_N	PCIE lane0 Transmit	OUT	48	/		
49	A_PCIE0_TX0_P			50	A_PCIE0_RST	PCIE Reset	3.3V

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51	GND			52	A_PCIE0_CLKREQ	PCIE Clock Request	IN 3.3V
53	A_PCIE2_CLK_N	PCIE ref CLK	OUT	54	PCIE_WAKE	PCIE wake	IN
55	A_PCIE2_CLK_P			56	/		
57	GND			58	/		
59	KEY		Unused	60	KEY		Unused
61				62			
63				64			
65				66			
67	/			68	/		
69	/			70	P3V3_M2M		
71	GND			72	P3V3_M2M		
73	GND			74	P3V3_M2M		
75	GND						

5 Board Expansion IF Connectors

Kalos Carrier board supports several expansion headers and connectors that have custom pinouts. The following lists the headers and connectors inform.

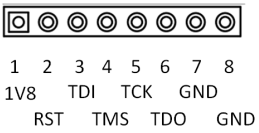
- *Kalos Module Connector, 260-pin, SO-DIMM, 0.5 mm pitch*
- *Camera Connectors (x2), 22 position, Flex Connector, 0.5 mm pitch*
- *Display Connectors (x1), 22 position, Flex Connector, 0.5 mm pitch*
- *40-Pin Expansion Header, 2x20, 2 mm pitch*
- *20-Pin Expansion Header, 1x20, 2 mm pitch*
- *Fan Connector, 4-pin, 1.25 mm pitch*
- *DC Power Jack*

5.1 Carrier Board 40+20 pin expansion header

The Carrier Board includes a 40-pin (2x20, 2 mm pitch) J12 ,a 20-pin (1x20, 2mm pitch)expansion header (J19) and a 8-pin JTAG header(J25). The expansion connector includes various control interfaces including:

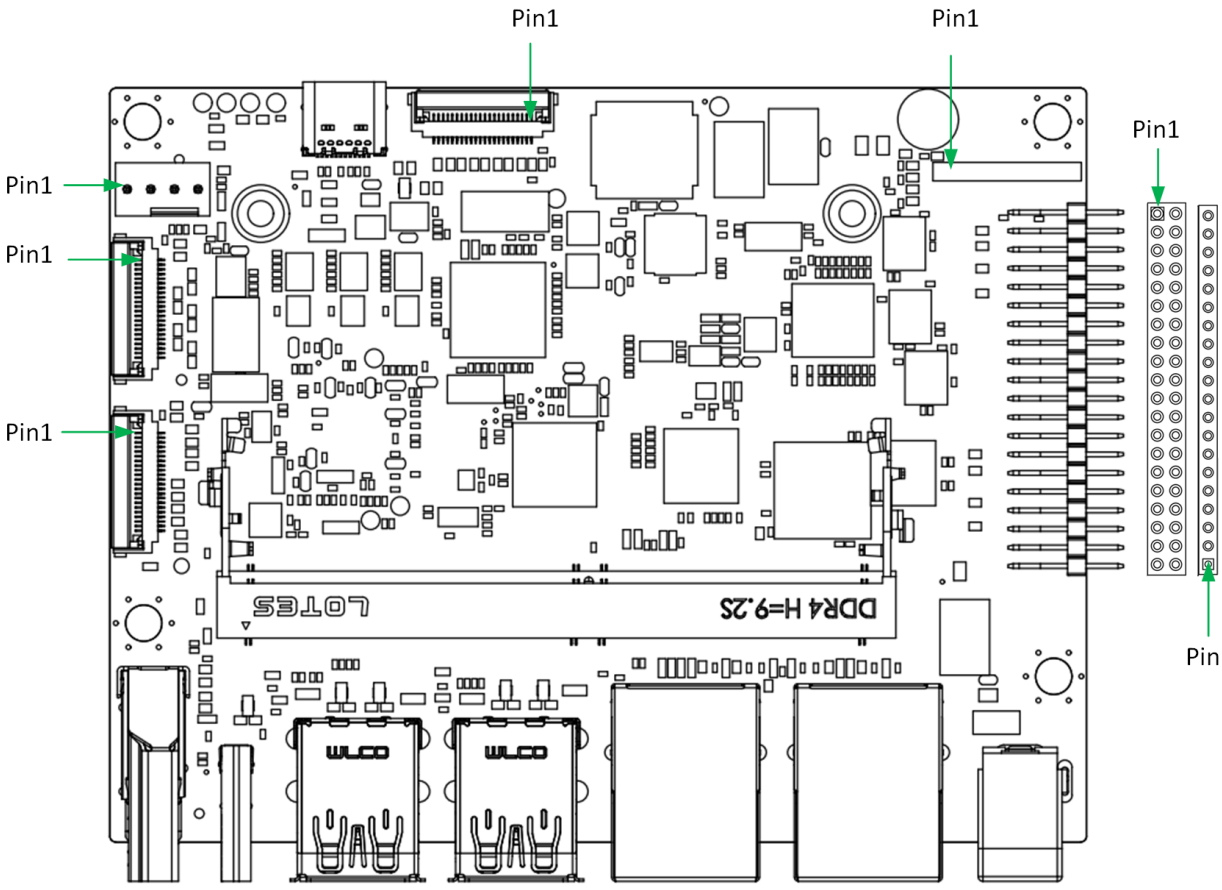
- *3xCAN-FD*
- *2xSPI*
- *2xUART*
- *4xPWM*
- *2xI2S*
- *8xI2C*
- *5xGPIO(For JTAG)*
- *3V3*
- *5V*
- *1V8*

Seeed



VDD_3V3_SYS	1		2	VDD_5V_GPIO		20	UART1_RTS_LS
I2C4_PCA9548_SDA3	3		4	VDD_5V_GPIO		19	UART1_TXD_LS
I2C4_PCA9548_SCL3	5		6	GND		18	UART1_RXD_LS
I2C4_PCA9548_SDA4	7		8	UART3_TXD_LS		17	UART1_CTS_LS
I2C4_PCA9548_SCL4	9		10	UART3_RXD_LS		16	CAN1_FD_L_A
UART3_RTS_LS	11		12	I2S0_CK_LS		15	CAN1_FD_H_A
SPI1_SCLK_LS	13		14	CAN0_FD_L_A		14	VDD_3V3_SYS
PWM3_LS	15		16	CAN0_FD_H_A		13	CAN2_FD_L_A
VDD_3V3_SYS	17		18	SPI1_MISO_LS		12	CAN2_FD_H_A
SPI0_MOSI_LS	19		20	A_I2S0_MCK_L		11	GND
SPI0_MISO_LS	21		22	SPI1_MISO_LS		10	PWM0_LS
SPI0_SCK_LS	23		24	SPI0_CS0_LS		9	PWM1_LS
I2S1_CL_LS	25		26	PWM2_LS		8	I2C4_PCA9548_SDA7
ID_I2C_SDA	27		28	ID_I2C_SCL		7	I2C4_PCA9548_SCL7
I2S1_SDIN_LS	29		30	PWR_BTN_3V3		6	I2C4_PCA9548_SDA0
I2S1_LRCK_LS	31		32	I2C4_PCA9548_SDA5		5	I2C4_PCA9548_SCL0
I2S1_SDOUT_LS	33		34	I2C4_PCA9548_SCL5		4	I2C4_PCA9548_SDA1
I2S0_LRCK_LS	35		36	UART3_CTS_LS		3	I2C4_PCA9548_SCL1
SPI1_MOSI_LS	37		38	I2S0_SDIN_LS		2	I2C4_PCA9548_SDA2
GND	39		40	I2S0_SDOUT_LS		1	I2C4_PCA9548_SCL2

Figure 6 Kalos Carrier Board Expansion Header Connectors
(J25 / J12 / J19)



5.2 Carrier Board CSI MIPI 22 pin Connector

The Carrier Board includes two 22-position flex (0.5 mm pitch) camera connectors (J20 and J21). The connector used on the carrier board Molex Part #54548-2272.

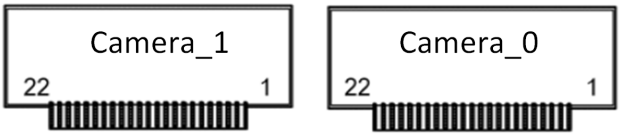


Figure 7 Kalos Carrier Board Camera MIPI Connectors Pin1 Placement

Table 8 Carrier Board CSI pin assignments - J20

Pin NO.	Pin Name	——	Usage/Description	Type/Dir
1	VDD_3V3_SYS		VDD_3.3V	
2	CAM0_I2C_SDA		CAM_I2C_SDA	Bidir
3	CAM0_I2C_SCL		CAM_I2C_SCL	OUTPUT
4	GND		GND	
5	CAM0_MCLK		CAM_MCLK	OUTPUT
6	CAM0_PWDN_LS		CAM_PWDN_LS	OUTPUT
7	GND		GND	
8	A_MIPI1_DATA3_P		CSI_D3P	INPUT
9	A_MIPI1_DATA3_N		CSI_D3N	INPUT
10	GND		GND	
11	A_MIPI1_DATA2_P		CSI_D2P	INPUT
12	A_MIPI1_DATA2_N		CSI_D2N	INPUT
13	GND		GND	
14	A_MIPI1_CLK_P		CSI_CLKP	INPUT
15	A_MIPI1_CLK_N		CSI_CLKN	INPUT
16	GND		GND	
17	A_MIPI1_DATA1_P		CSI_D1P	INPUT
18	A_MIPI1_DATA1_N		CSI_D1N	INPUT
19	GND		GND	
20	A_MIPI1_DATA0_P		CSI_D0P	INPUT
21	A_MIPI1_DATA0_N		CSI_D0N	INPUT
22	GND		GND	

Table 9 Carrier Board CSI pin assignments - J21

Pin NO.	Pin Name	——	Usage/Description	Type/Dir
1	VDD_3V3_SYS		VDD_3.3V	
2	CAM1_I2C_SDA		CAM_I2C_SDA	Bidir
3	CAM1_I2C_SCL		CAM_I2C_SCL	OUTPUT
4	GND		GND	
5	CAM1_MCLK		CAM_MCLK	OUTPUT
6	CAM1_PWDN_LS		CAM_PWDN_LS	OUTPUT
7	GND		GND	
8	A_MIPI0_DATA3_P		CSI_D3P	INPUT
9	A_MIPI0_DATA3_N		CSI_D3N	INPUT
10	GND		GND	

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11	A_MIPI0_DATA2_P		CSI_D2P	INPUT
12	A_MIPI0_DATA2_N		CSI_D2N	INPUT
13	GND		GND	
14	A_MIPI0_CLK_P		CSI_CLKP	INPUT
15	A_MIPI0_CLK_N		CSI_CLKN	INPUT
16	GND		GND	
17	A_MIPI0_DATA1_P		CSI_D1P	INPUT
18	A_MIPI0_DATA1_N		CSI_D1N	INPUT
19	GND		GND	
20	A_MIPI0_DATA0_P		CSI_D0P	INPUT
21	A_MIPI0_DATA0_N		CSI_D0N	INPUT
22	GND		GND	

5.3 Carrier Board DSI MIPI 22 pin Connector

The Carrier Board provides a 22-position flex (0.5 mm pitch) connector for MIPI Display (J3). The connector used on the carrier board is CF20221D0R0-NH.

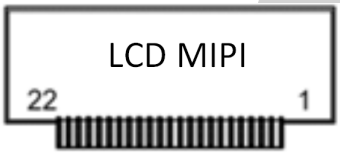


Figure 8 Kalos Carrier Board Display MIPI Connector Pin1 Placement

Table 10 Carrier Board DSI pin assignments - J3

Pin NO.	**Pin Name**	——	**Usage/Description**	**Type/Dir**
1	VCC_3V3		3.3V Power Supply	
2	LVDS_TX_SDA		SDA	
3	LVDS_TX_SCL		SCL	
4	GND		GND	
5	/		NC	
6	/		NC	
7	GND		GND	
8	MIPI_A_D3+/LVDS_A_D0+		DSI_D3P	
9	MIPI_A_D3-/LVDS_A_D0-		DSI_D3N	
10	GND			
11	MIPI_A_D2+/LVDS_A_D1+		DSI_D2P	
12	MIPI_A_D2-/LVDS_A_D1-		DSI_D2N	
13	GND			
14	MIPI_A_CLK+/LVDS_A_D2+		DSI_CLKP	
15	MIPI_A_CLK-/LVDS_A_D2-		DSI_CLKN	
16	GND			
17	MIPI_A_D1+/LVDS_A_CLK+		DSI_D1P	
18	MIPI_A_D1-/LVDS_A_CLK-		DSI_D1N	
19	GND			
20	MIPI_A_D0+/LVDS_A_D3+		DSI_D0P	

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21	MIPI_A_D0-/LVDS_A_D3-		DSI_D0N	
22	GND			

5.4 Fan Connector

Table 11 The Carrier board includes a 4-pin fan header (J32).

Pin NO.	**Pin Name**	——	**Usage/Description**	**Type/Dir**
1	GND		GND	
2	FAN_PWR			
3	FAN_SPEED_1		FAN TACH	
4	PWM3_FAN/ MCU_PWM_FAN		PWM FAN CONTROL	

5.5 DC Power Jack

The Kalos carrier board uses a DC power jack (J16) to bring in the power from the included DC power supply. The mating barrel jack connector dimensions.

Length: 11 +/- 0.5 mm
Diameter: 5.5 mm
Diameter2: 2.5 mm jack pin
The center pin is positive (+V)
Max current supported is 10A

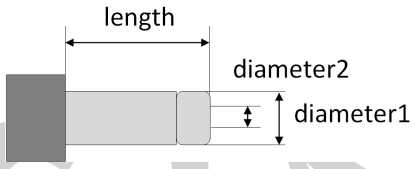


Figure 9 DC Plug (DC-2.5-PU)

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6 Power Supplies

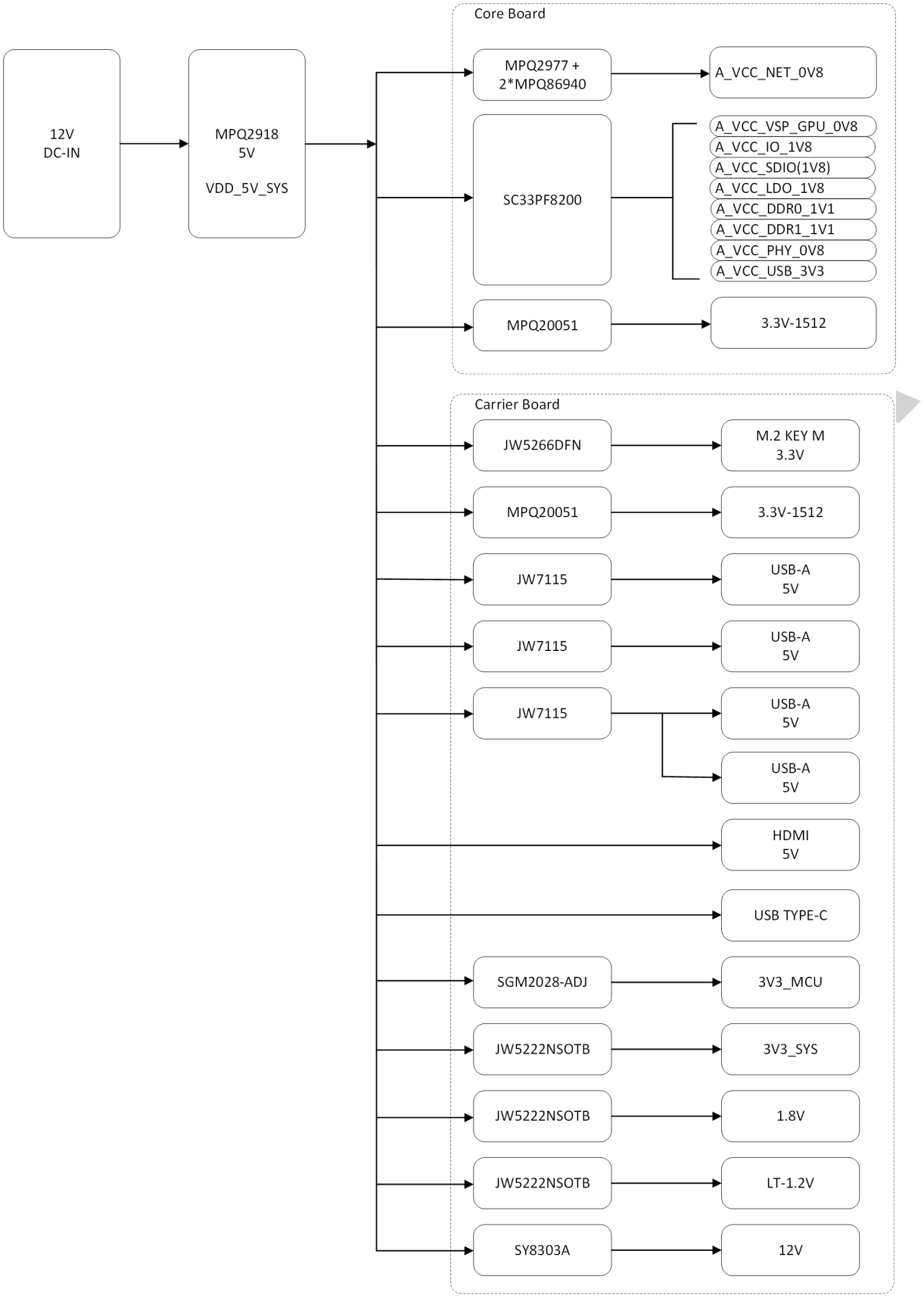


Figure 10 Kalos Core Board & Carrier Board Power Diagram

The following tables show the allocation of supplies to the connectors on the carrier board and current Capabilities.

Table 12 Supply Current Capabilities per Connector per Supply

Power	**Connector**	**Typical Voltage(V)**	**Max Current (A)**
VBUS	USB-A	5	1
VBUS	USB-C	5	1
HDMI	HDMI CONNECTOR	5	1
VDD_5V_GPIO	40 PIN HEADER	5	1
P3V3_M2M	M.2 KEY M	3.3	2.5
VDD_3V3_SYS	DSI/CSI CONNECTOR	3.3	0.5
	40/20 PIN HEADER		
VDD_1V8	/	1.8	0.5

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7 Device Ordering Information

DX-3000-D	Module Form Factor : SO-DIMM	
DX-3000-D008	Memory Config: 8GB DRAM	

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